

Mathematical Analysis II

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Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

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Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.33.1 and Mathlib v4.33.1 (revision [0df444a360ea](#)).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.
- **Prerequisites.** Starting assumptions and accepted external results are specific to this textbook.

Chapter 1

Sequences and Series of Functions

1.1 Discussion of Main Problem

1.2 Uniform Convergence

1.3 Uniform Convergence and Continuity

1.4 Uniform Convergence and Integration

1.5 Uniform Convergence and Differentiation

1.6 Equicontinuous Families of Functions

1.7 The Stone-Weierstrass Theorem

Chapter 2

Some Special Functions

2.1 Power Series

2.2 The Exponential and Logarithmic Functions

2.3 The Trigonometric Functions

2.4 The Algebraic Completeness of the Complex Field

2.5 Fourier Series

2.6 The Gamma Function

Chapter 3

Functions of Several Variables

3.1 Linear Algebra Prerequisites

This chapter assumes finite-dimensional linear algebra: bases, dimension, linear maps and their matrix representations, rank–nullity, inner products, orthogonalization, and norms. These results are used as prerequisites rather than developed again. We also assume equivalence of norms in finite dimensions and continuity of linear maps.

We write $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ for a linear map and use the same letter for its matrix in the standard bases. Its rank is the dimension of its image. Unless specified otherwise, vector norms are Euclidean. The operator norm is

$$\|A\| = \sup_{\|x\| \leq 1} \|Ax\|.$$

We use the standard estimate $\|Ax\| \leq \|A\|\|x\|$. These conventions let us describe derivatives as linear maps. The analytic arguments establishing differentiability and the inverse, implicit, and rank theorems belong to this chapter and are not part of the linear algebra assumed here.

3.2 Differentiation

3.3 The Contraction Principle

3.4 The Inverse Function Theorem

3.5 The Implicit Function Theorem

3.6 The Rank Theorem

3.7 Determinant Notation

Determinants are assumed from finite-dimensional linear algebra. For a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\det A$ denotes the determinant of its matrix in the standard basis. We use multilinearity, alternation, $\det I = 1$, and

$$\det(AB) = \det A \det B, \quad A \text{ is invertible} \iff \det A \neq 0.$$

No reconstruction of determinant theory is needed here. In multivariable analysis, the determinant of the derivative expresses the nonsingularity condition. Its role in change of variables is developed with integration; the change-of-variables theorem is not an assumed linear algebra result.

3.8 Derivatives of Higher Order

3.9 Differentiation of Integrals

Chapter 4

Integration of Differential Forms

4.1 Integration

4.2 Primitive Mappings

4.3 Partitions of Unity

4.4 Change of Variables

4.5 Differential Forms

4.6 Simplexes and Chains

4.7 Stokes' Theorem

4.8 Closed Forms and Exact Forms

4.9 Vector Analysis

Chapter 5

The Lebesgue Theory

5.1 Set Functions

5.2 Construction of the Lebesgue Measure

5.3 Measure Spaces

5.4 Measurable Functions

5.5 Simple Functions

5.6 Integration

5.7 Comparison with the Riemann Integral

5.8 Integration of Complex Functions

5.9 Functions of Class $\mathcal{L}^2$

Bibliography

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